Fiscal Year:	FY 2008	**Task Last Updated:**	FY 09/25/2008
PI Name:	Masson, Patrick H. Ph.D.		
Project Title:	Molecular Genetics of Root Thigmoresponsiveness in Arabidopsis thaliana		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	phmasson@wisc.edu	**Fax:**	FY 608-262-2976
PI Organization Type:	UNIVERSITY	**Phone:**	608-265-2312
Organization Name:	University of Wisconsin Madison		
PI Address 1:	Laboratory of Genetics (Room 3262)		
PI Address 2:	425G Henry Mall		
PI Web Page:	https://		
City:	Madison	**State:**	WI
Zip Code:	53706-1580	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
Start Date:	07/01/2004	**End Date:**	06/30/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: NCE through 6/30/2008 per A. Chu/ARC (6/07) 8/06--funding reinstated through 6/30/2007, per J. Cohen (HQ)		
Key Personnel Changes/Previous PI:	Masson, Patrick		
COI Name (Institution):			
Grant/Contract No.:	NNA04CC71G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Directional cues within the environment (gravity, touch, light etc) guide plant organs' growth by differential regulation of cell expansion. These tropic movements are coupled to endogenous oscillating movements (nutations) whose origin remains poorly understood. Together, these movements allow plant organs to position themselves for optimal performance. To better understand the mechanisms that govern integrated growth responses to multiple stimuli, we have sought and characterized Arabidopsis thaliana mutations that affect root waving and skewing on tilted hard agar surfaces. Such movements result from a combination of gravity, touch, hydrotropic and other surface-derived stimuli, as well as circumnutations. One of these mutations, named wvd2-1, was shown to alter spiral root growth on agar surfaces and anisotropic cell expansion. The WVD2 protein appears to modulate the arrangement of cortical microtubules in expanding cells. This proposal is aimed at understanding how WVD2 and other related proteins modulate anisotropic cell expansion and root growth behavior. This will be accomplished by identifying and characterizing proteins that regulate or modulate WVD2 function, and by characterizing seven other WVD2-related proteins in Arabidopsis. Plants constitute an essential component of Advanced Life Support Systems for long-term space-exploration missions. However, during space flight, their growth will be guided by cues other than gravity, including nutations. Therefore, we need to better understand some of the mechanisms that will contribute to guiding organ growth under these conditions, a primary objective of this proposal. Also, there is limited space available for plant growth in spacecrafts. WVD2 may provide a good tool to design reversible engineering strategies that would confer a shorter stature to a broad range of plants, yet would not alter seed production or plant's ability to perceive gravity at their final destination. Hence, a thorough characterization of WVD2-dependent pathways is of direct relevance to the Space Biology Program. The objective of this proposal is to characterize the molecular mechanisms that allow WVD2 and related WDL proteins to control root growth behavior in response to mechanical stimuli, anisotropic cell expansion and plant morphogenesis. Our long-term goal is to use the information that will derive from this project to engineer shorter and stockier plants than wild type, which can still produce normal levels of seeds, but are better suited for growth in the limited confines of bioregenerative life support systems during long-term space exploration missions, or that better resist to lodging by wind and rain on Earth or other planets. The WVD2 and WDL genes appear to function not only as key regulators of root growth behavior in response to mechanical stimuli, but also as major determinants in the control of anisotropic cell expansion in all organs of the plant. Therefore, a better understanding of these genes is likely to provide important new insights into how mechanical information in the environment controls cell expansion, and how this affects overall plant morphogenesis, a process of fundamental importance for agricultural production and plant adaptation to its environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Root-growth behavior conditions the ability of plants to populate and grow in very heterogenous and diverse environments. Hence, a better understanding of the molecular mechanisms that govern this process is likely to provide the tools needed to engineer plants that are better able to colonize extreme environments on Earth, and on other planets. At the same time, controlled anisotropic cell expansion is an essential component of plant morphogenesis and behavior in response to directional environmental stimuli. Our studies have allowed the identification of an important new dominant regulator of this process. Because over-expression of WVD2 results in shorter plants with thicker organs, it may be possible to use this engineering strategy to generate crop plants that are more resistant to lodging by wind and rain, as well as plants that are better suited for utilization in spacecrafts, where limited amounts of space are available for plant growth.
Task Progress:	The research supported by this grant was aimed at characterizing the roles of a protein, named WVD2, and of related WVD2-Like (WDL) proteins, in the control of root growth behavior on hard-agar surfaces and of anisotropic cell expansion in the model plant Arabidopsis thaliana. Using a combination of Genetics, Cell Biology and Biochemistry, we have shown that these proteins modulate cell expansion in plant organs by interacting with, and modulating the bundling of, cortical microtubules within those cells. These bundling effects result in alterations of the dynamic instability of the microtubules, with probable effects on the patterns of wall-components deposition in expanding cells and subsequent alterations in the patterns of expansion and growth behavior. We also found that at least two of these proteins, WVD2 and WDL5, interact with, and are phosphorylated by, the kinase domain of the receptor to a group of plant hormones, the brassinosteroids, which were previously shown to modulate cell expansion. These exciting in vitro results suggest possible direct interactions between the brassinosteroid receptor and proteins that directly contribute to the rearrangement of the cortical microtubules in expanding cells. Importantly, other WDL proteins seem to also regulate anisotropic cell expansion and growth behavior in plants, showing evidence of both functional redundancy and functional specialization within the family. Our studies of wvd2-1 have also uncovered an unexpected enhancement of thigmomorphogenesis, a process that allows plants exposed to repeated mechano-stimuli to grow more slowly and expand radially relative to unstimulated controls. This observation is compatible with a role for the microtubules in mechano-transduction in plants. Hence, wvd2-1 constitutes a sensitized background that may prove useful in genetic investigations of touch sensing and mechano-transduction, a process of critical relevance to the Fundamental Space Biology Program at NASA. Our investigations on the root growth behavior also uncovered a role for the secretory pathway in the control of anisotropic cell expansion, suggesting the potential involvement of wall- or plasma-membrane-associated proteins in regulating this process. Future work will be directed at uncovering these potentially key regulators of plant growth and development.
Bibliography Type:	Description: (Last Updated: 07/03/2024)
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